Topic:

Network Security

Subject: Computer Science

Research Question:

“How Vulnerable is a Home Router? A Penetration Testing Study Using Wireshark”

Word Count: 3867

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Introduction

In today's hyperconnected digital world, networks have become a foundational component of our lives. We can work remotely, study online, and manage our finances. As network usage grows rapidly, the risk of personal information theft increases accordingly.1.1 million people in the United States had their personal data stolen, and the number of individuals affected is growing(Experian, 2024). This Extended Essay investigates one of the most relevant topics in modern cybersecurity and aims to answer the following question: "How Vulnerable is a Home Router?" This question is under the topics of Computer Science as it is focused on the decisions regarding router's firmware that are creating weaknesses in the network. This investigation is not focused on user behaviour or general cybercrime, it is focused on weaknesses and solutions of a router with a default settings.

Understanding these concepts provides a necessary foundation for the following background section, which will explore key principles relevant to assessing home router vulnerability.

(Figure 1.1 - Wireshark**)**

Using **Wireshark** (figure 1.1), a widely used open-source packet analyzer, this investigation simulates a basic penetration test to provide insight into what kinds of data might be exposed to attackers on an unsecured network.

This study aims not only to uncover potential weaknesses in a common home setup but also to suggest basic, practical security practices that everyday users can implement, such as updating firmware or enabling security settings not offered by default by the internet provider.

Background

Network Protocols

The most used internet protocols are

Protocol	Full Name	Purpose	Typical Port(s)
TCP/IP	Transmission Control Protocol / Internet Protocol	Core protocol suite for internet communication.	—
UDP	User Datagram Protocol	Used in DNS communications and fast, connectionless transfers.	—
HTTP	Hypertext Transfer Protocol	Used in the login test; shows vulnerability due to lack of encryption.	80
HTTPS	Hypertext Transfer Protocol Secure	Mentioned as the secure alternative to HTTP.	443
DNS	Domain Name System	Used to resolve website names (e.g., youtube.com) to IP addresses.	53 (TCP/UDP)
ARP	Address Resolution Protocol	Used in ARP test to map IP addresses to MAC addresses.	No port (data link layer)

(Table 1.1 - Network protocols with their explanations**)**

(Auvik Networks, 2025; TechTarget, 2025; Iana, 2025).

How home Networks work

A home network connects devices like computers, smartphones, tablets, smart TVs, printers, and game consoles within a residence, enabling communication and resource sharing, including internet access and file sharing.

Core Components

- Access Point: Manages wireless connections. They are usually integrated into **routers.**
- Router: The central network hub, managing both wired (Ethernet) and wireless (Wi-Fi) connections. Modern routers often integrate wireless access point and **switch** functions, connecting devices internally and to the internet.
- Switch: An Optional device used to increase the number of wired LAN ports.
- Modem: Connects the home network to the Internet Service Provider (ISP). Modern setups often combine a modem and a router into a single device.

Network Types

Type	Full Name	Explanation
LAN	Local Area Network	Connects local devices in a small area.
WLAN	Wireless LAN	Wi-Fi network in a home or office.
VPN	Virtual Private Network	Encrypted connection over public network.
WAN	Wide Area Network	Connects home network to the Internet.

(Table 1.2 - Network Types**)**

Wireshark

Wireshark is a free and widely used network protocol analyzer that captures and analyzes network traffic (Wireshark Foundation, 2025). It allows users to view network packets as they traverse a network, showing details such as source and destination addresses, protocols, and packet contents(Wireshark Foundation, 2025). Professionals

and security researchers use Wireshark to troubleshoot network issues(Wireshark Foundation, 2025).

Detailed Features and Functions of Wireshark

Feature / Use	Description
Packet Capture	Captured live traffic.
Protocol Support	Used TCP, UDP, HTTP, DNS, ARP.
Filtering	Isolated HTTP, DNS, ARP traffic.
Packet Details	Viewed headers and payloads.
Reconnaissance	Found IP/MAC via ARP, DNS.
Unencrypted Data	Exposed login via HTTP.
Credential Analysis	Extracted HTTP form data.
Interface Selection	Picked the correct Wi-Fi adapter.
Capture Filters	Applied DNS/HTTP/ARP filters.
Live Capture & Stop	Captured and stopped tests.
Analysis	Inspected packet content.
Encrypted Traffic	Not readable without keys.

Legal Use	Done on own network.

(Table 1.3 - Wireshark functions**)**

Methodology

This study has used a structured penetration testing style guidance regarding passive packet capture. All the tests shown in this investigation were done only on the researcher's home network using only the researcher's routers with default configurations. This decision was made to practically emphasize the real world examples of non-experts operating their devices. During this research the only tool for testing was Wireshark. It was selected for this study because it has the ability to inspect packets on different levels of OSI model, filter specific data types such as HTTP, DNS and ARP, and rebuild payloads without distracting the live traffic of the network. The test was done using three user devices that were connected to a single router with model G-97RGT. This investigation focused only on passive observations to ensure that any discovered weaknesses operated normally and not after any manipulations.

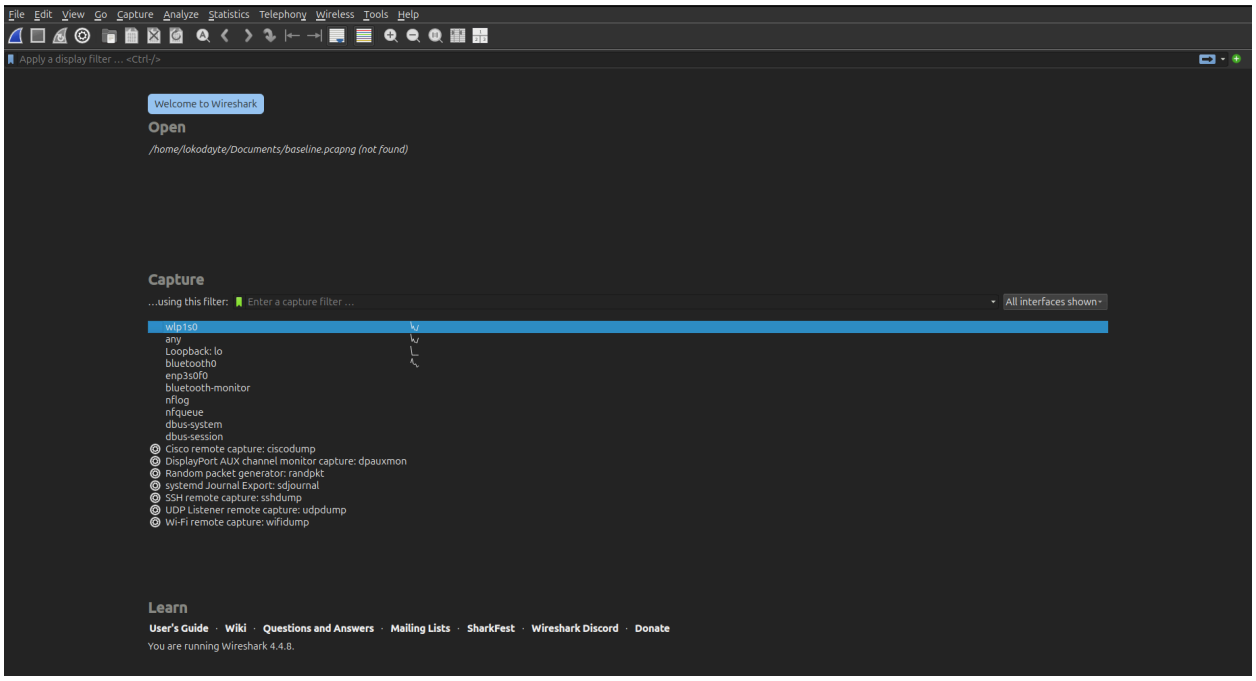
Three tests were chosen: an HTTP login test, a DNS lookup test, and an ARP observation. These represent realistic threats at the application, transport, and data link layers. The tests show how an unauthorized user without special access could exploit certain vulnerabilities. No active attacks, such as ARP spoofing or packet injection, were performed, and no custom firmware was used. This approach kept the focus on default router behavior and followed ethical guidelines.

Other tools like tcpdump and Nmap were also considered, but Wireshark was chosen because it can examine network data more closely and presents information in a way that is easier to understand. Each test was designed to answer the main research question, demonstrating that a typical home router cannot detect or stop these quiet threats. This enabled the identification of real security problems in regular home networks.

Practical Work

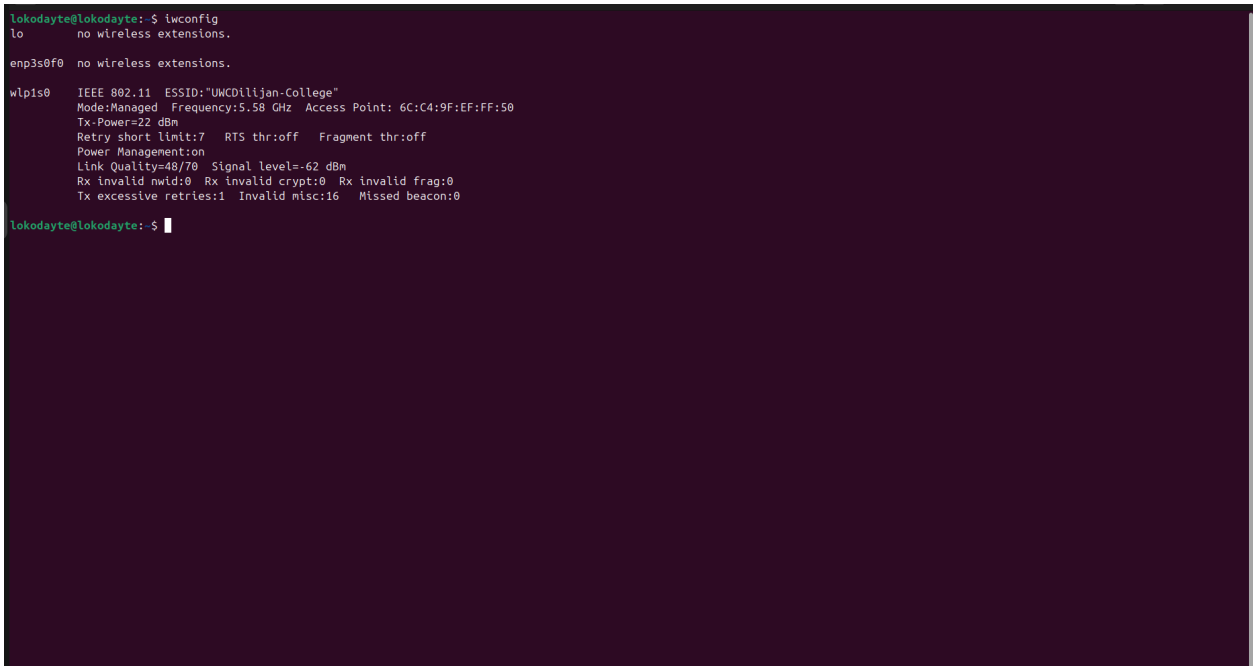
In this section the practical part of the investigation is presented, while using example testing machines that are publicly accessible. The analysis part is based on this practical component

Wireshark



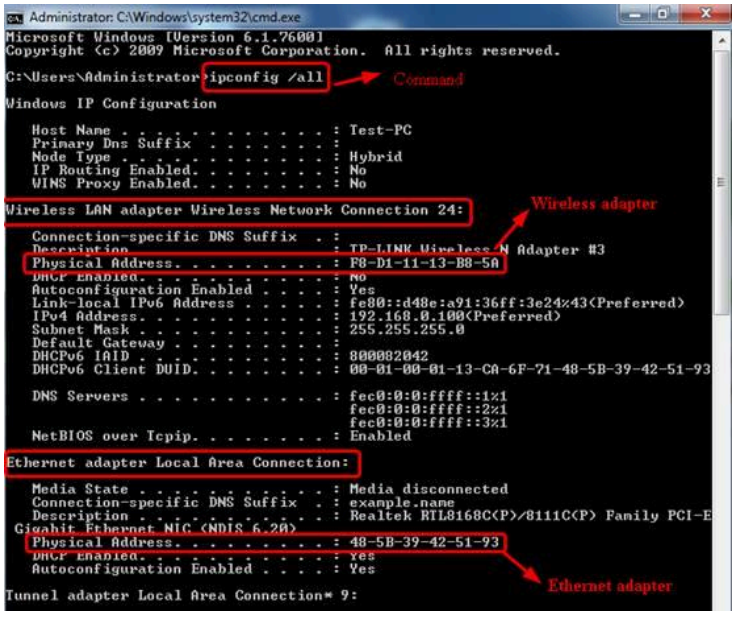
(Figure 1.2 - Wireshark bootup screen**)**

When we start Wireshark, we must select the correct network interface for packet capture. For internet testing, select the interface responsible for packet transmission. In Figure 1.2, it is wlp1s0. To identify the relevant network interface in Ubuntu, run the terminal command 'iwconfig' to display interface details as in Figure 1.3.



(Figure 1.3 - Ubuntu terminal**)**

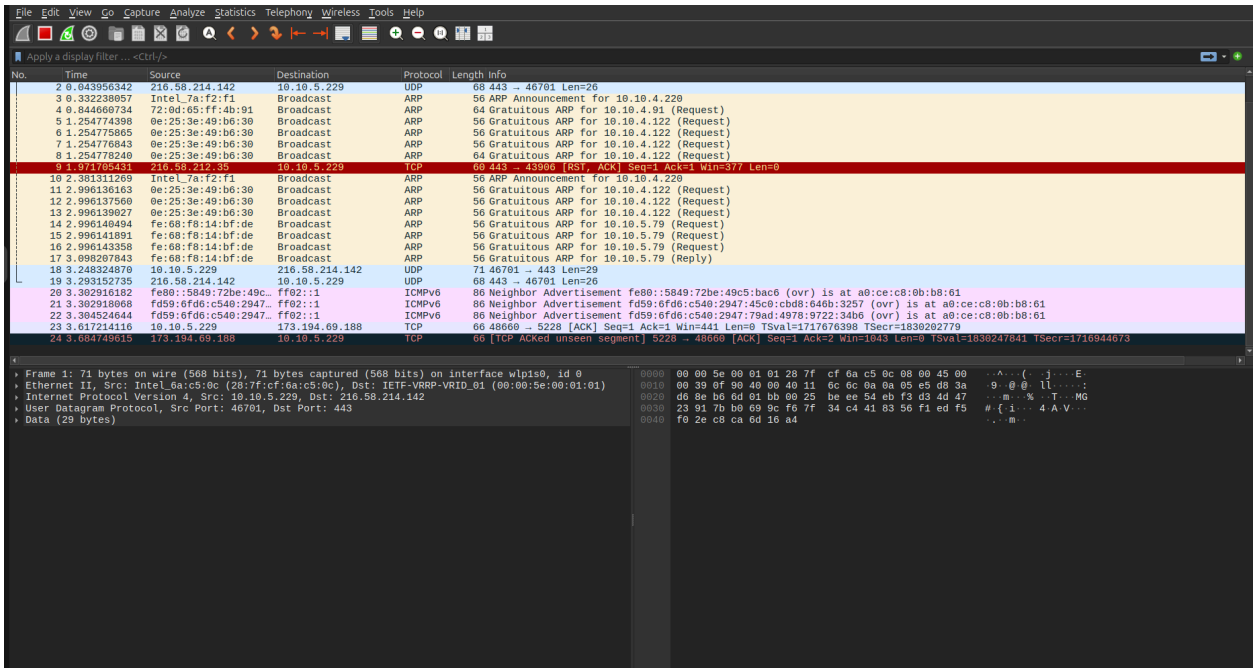
To view the network interface in Windows, open the Command Prompt and run the command. **ipconfig /all,** there we will see information about interfaces. The screen that will appear should look like Figure 1.4



(Figure 1.4 - Windows network interface in CMD**)**

The line above the Wireless **adapter** shows the name of our interface.

When the right interface is chosen, the screen should appear as in **Figure 1.5**.



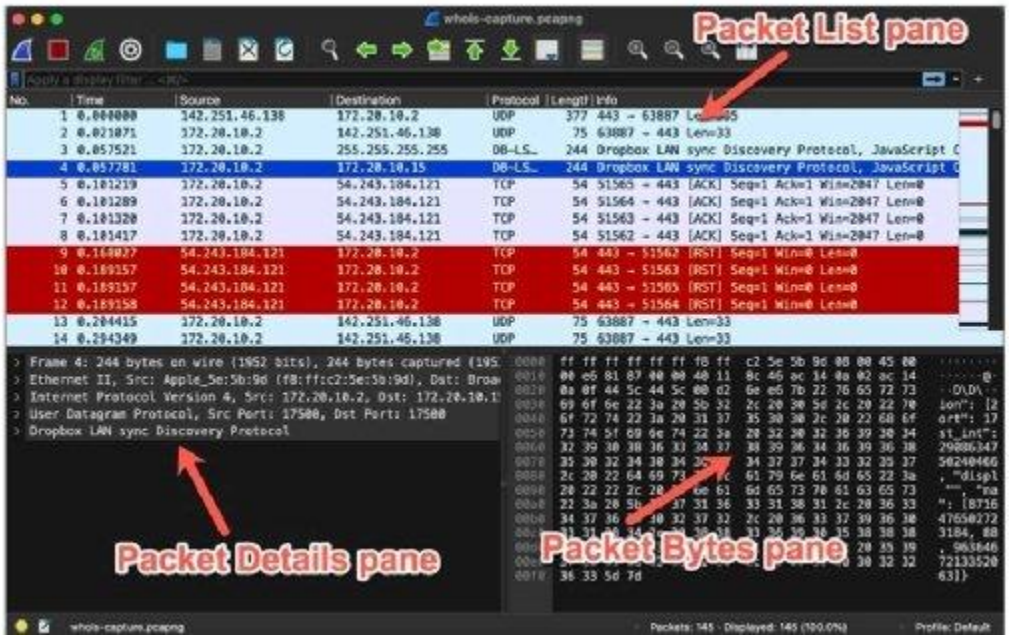
(Figure 1.5 - Wireshark screen for packet capturing**)**

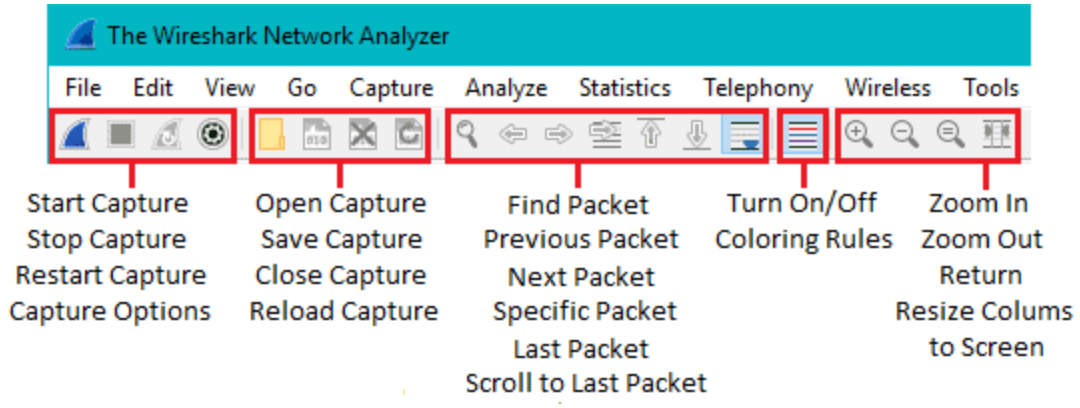
In the screen shown in **Figure 1.5,** we see network traffic. Here is information about the colors of the packets (each packet represents a formatted unit of data, typically a fragment of a larger message).

- **Light purple for TCP traffic**
- **Light blue for UDP traffic**
- **Black for packets with errors**

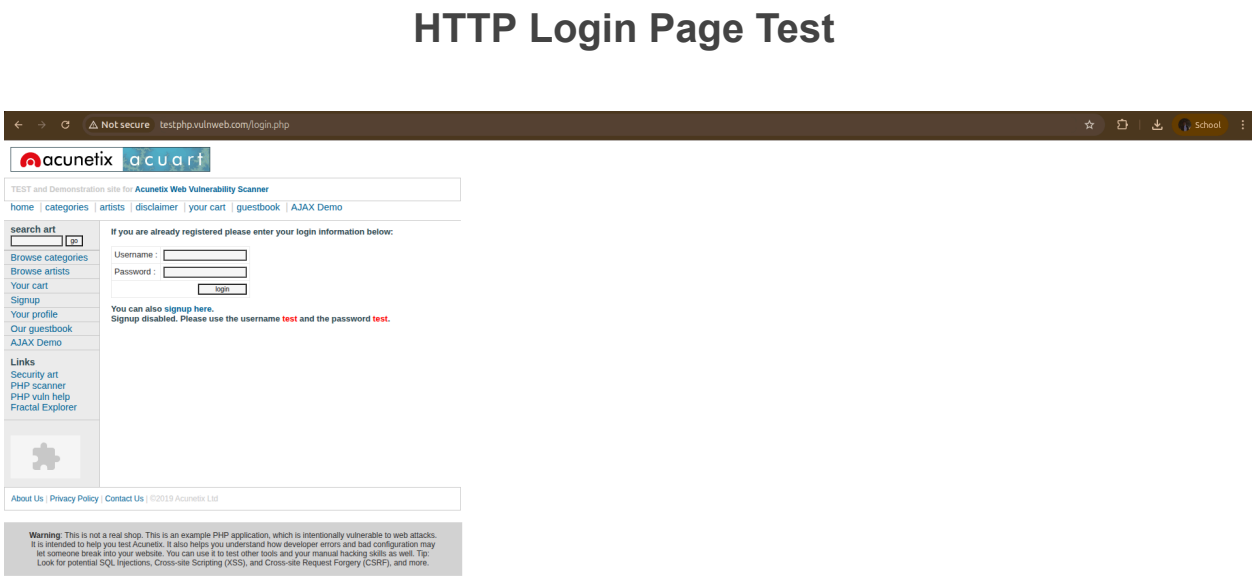
- **Light green for HTTP traffic**
- **Light yellow for Windows-specific traffic, such as SMB and NetBIOS**
- **Dark yellow for routing protocols**
- **Dark gray for TCP control packets like SYN, FIN, and ACK.**(Wireshark Foundation, 2025)

Figure 1.6 provides detailed information about **Wireshark** configurations.





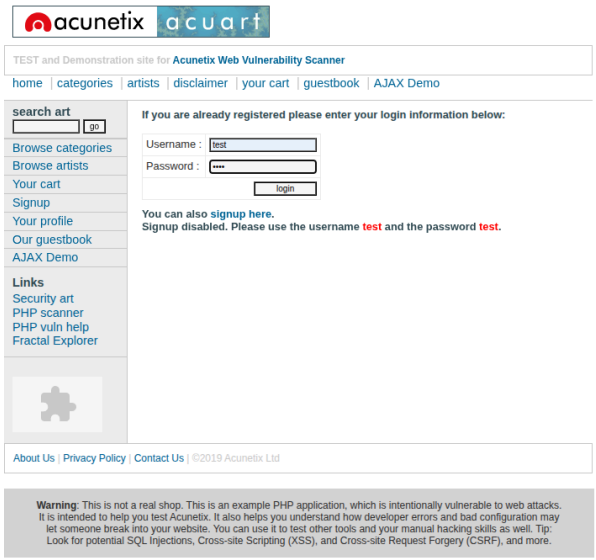
(Figure 1.6 - Wireshark user instructions**)**



(Figure 2.1 - HTTP website for tests**)**

The website shown in Figure 2.1 is for testing. To start with practical work, we open Wireshark and start it.

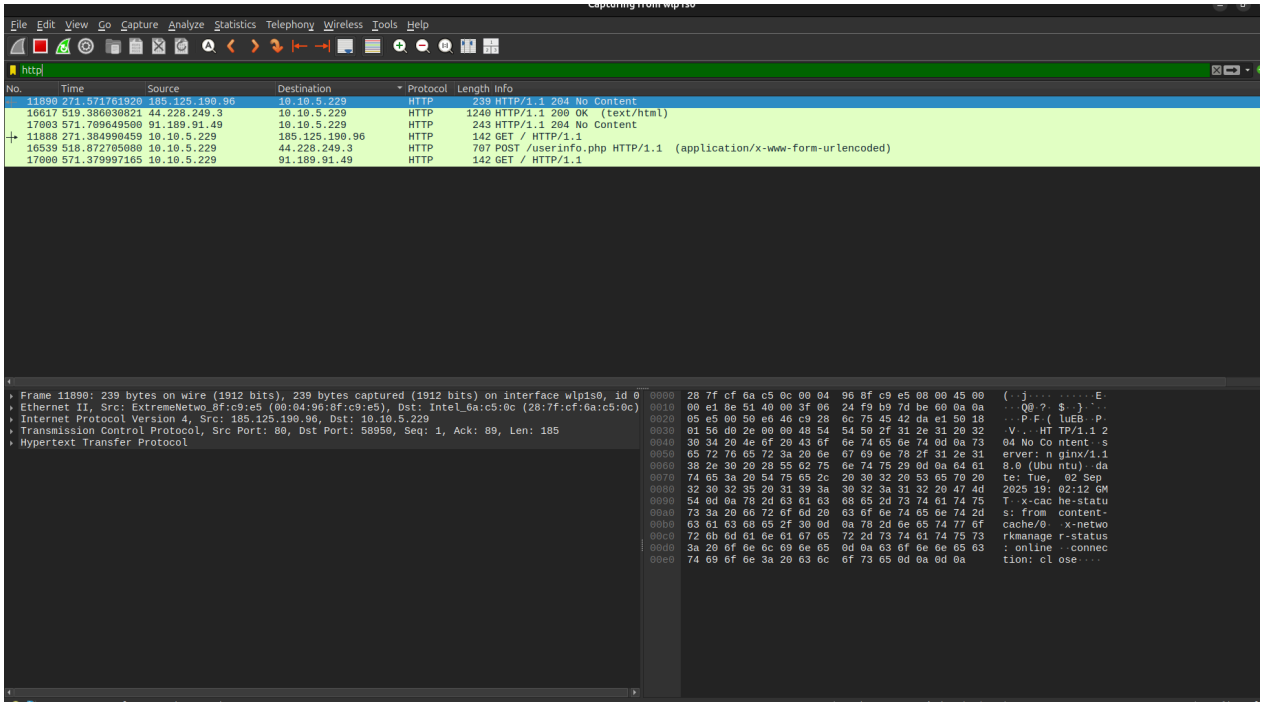
Fill in the login page and press the login button. As shown in Figure 2.2



(Figure 2.2 - HTTP login test page with written credentials**)**

As an example, in both fields, the word '*test*' was entered.

Since the website uses **HTTP,** we need to filter for **HTTP** in **Wireshark**.



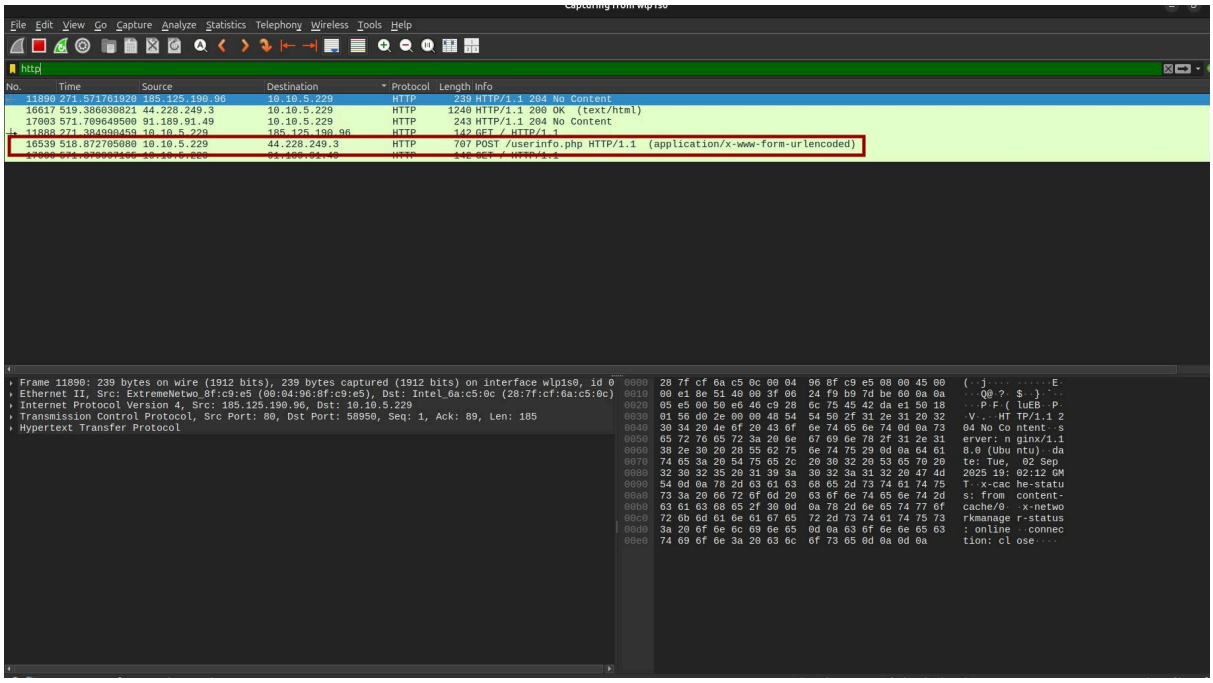
(Figure 2.3 - Filtered Wireshark packets**)**

Currently, we have 6 packets, each containing information about the HTTP destinations the laptop accessed. In one of them, there is information about the login page where the test was done.

In this case, the information that we need is located in this packet. We can tell the key indicators that this packet likely contains information about the login page are:

- The HTTP method is POST, commonly used for login forms to submit user credentials.
- The URL path "/userinfo.php" suggests it is related to user information or authentication.

- The content type "application/x-www-form-urlencoded" is the standard encoding for form submissions.



(Figure 2.4 - Wireshark packet selected**)**

- **Frame 16539: 707 bytes on wire (5656 bits), 707 bytes captured (5656 bits) on interface wlp1s0**

This is the packet number (16539), the total size of the packet in bytes and bits as it was transmitted on the network, and the number of bytes Wireshark captured for analysis. The interface name (wlp1s0) is the network device where the packet was captured.

- **Ethernet II, Src: Intel_6a:c5:0c (28:7f:cf:6a:c5:0c), Dst: IETF-VRRP-VRID_01 (00:00:5e:00:01:01)**

This is the data link layer header. It shows the source MAC address (Intel_6a:c5:0c), destination MAC address (IETF-VRRP-VRID_01), and the Ethernet protocol type. This layer handles communication on a local network segment.

- **Internet Protocol Version 4, Src: 10.10.5.229, Dst: 44.228.249.3**

This is the network layer header providing IPv4 addressing information, showing the source IP address (10.10.5.229) and destination IP address (44.228.249.3). It routes the packet across multiple networks to reach the destination.

- **Transmission Control Protocol, Src Port: 38538, Dst Port: 80, Seq: 1, Ack: 1, Len: 653**

This is the TCP transport-layer header. It includes the source port (38538) and destination port (80, the standard for HTTP), the sequence and acknowledgment numbers for reliable data transfer, and the length of the TCP payload (653 bytes). TCP ensures ordered and reliable communication.

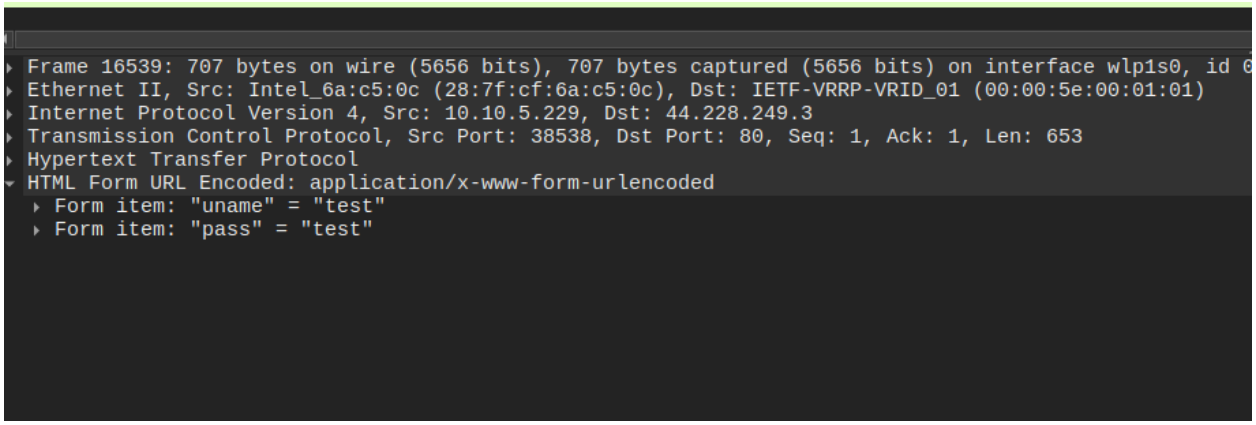
- **Hypertext Transfer Protocol**

This is the application layer showing the HTTP protocol data. It includes the POST method, indicating the submission of form data, the target resource (/userinfo.php), and the HTTP version 1.1.

- **HTML Form URL Encoded: application/x-www-form-urlencoded**

This specifies that the HTTP POST data is encoded as form data using key-value pairs, as in typical web form submissions. This encoding type sends form inputs as URL-encoded strings in the HTTP body.

If we open the last line **HTML Form…,** we will have access to the information that we were not supposed to have access to



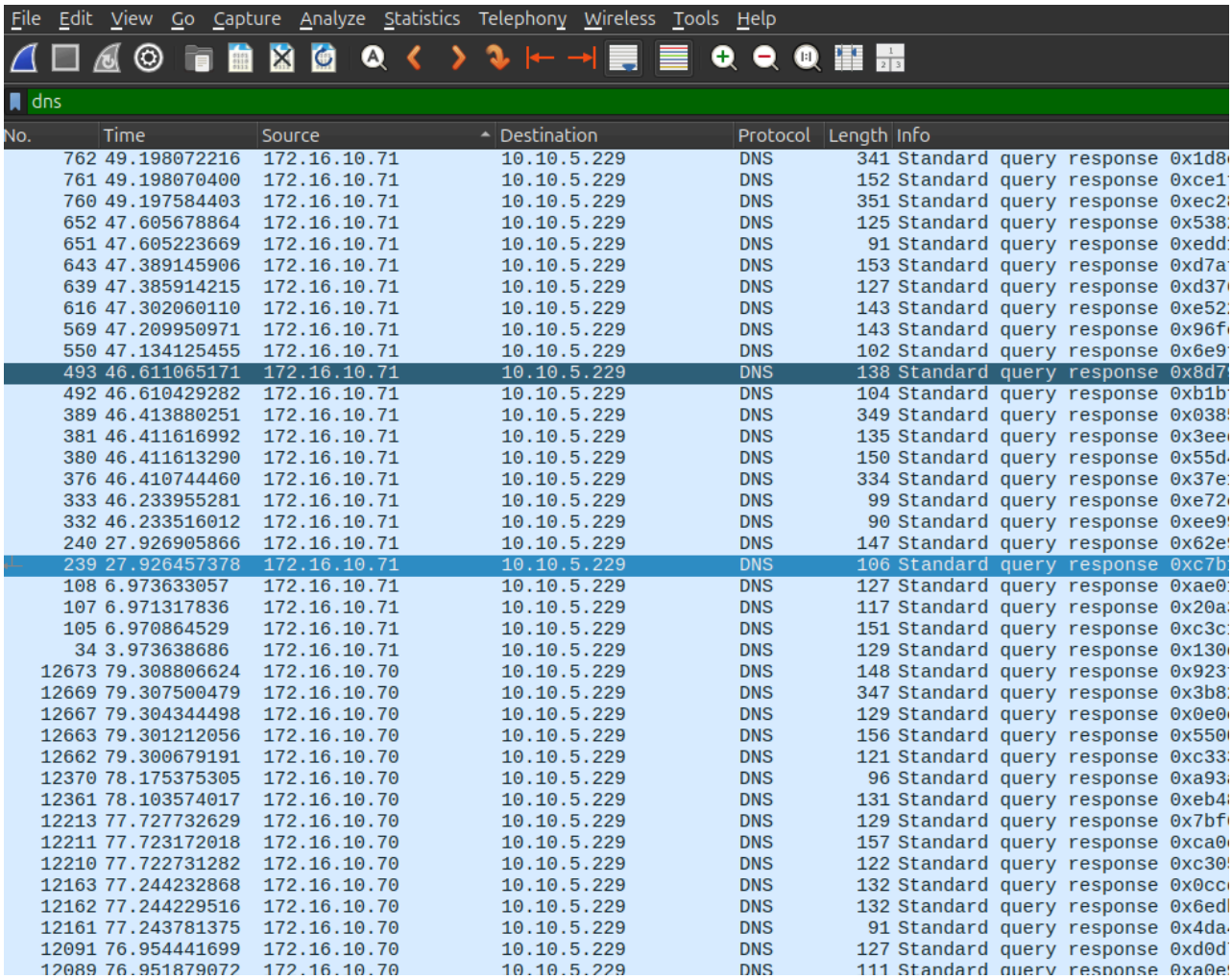
(Figure 2.5 - Information provided in Wireshark**)**

This test emphasizes that sensitive information, such as login credentials, can be transmitted in plain text over HTTP connections, which is insecure. The ability to capture data that should stay private highlights a significant vulnerability in home networks. It shows that without encryption, users are at risk of credential theft and unauthorized access.

DNS Lookup Test

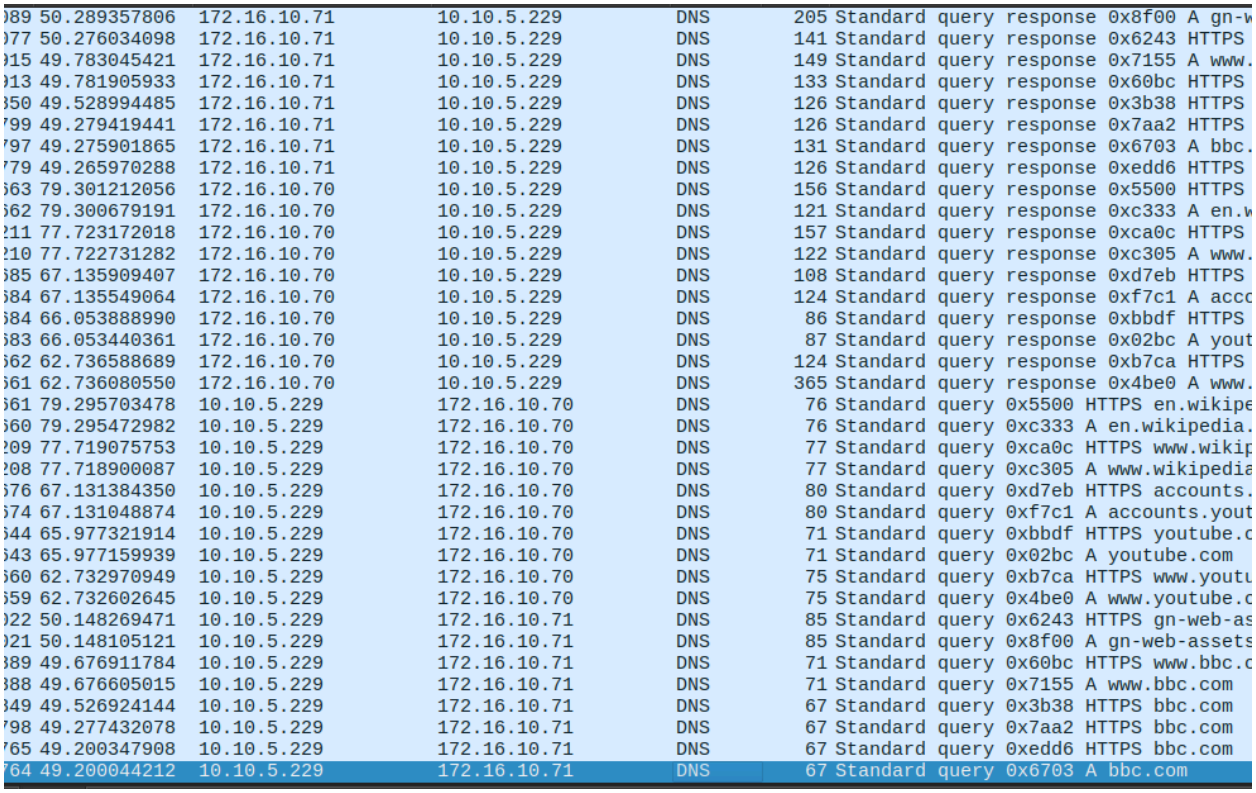
To start with this test, we are starting **Wireshark** the same way. While it captures data in the browser. While Wireshark captured data, a few websites were opened (bbc.com, wikipedia.org, youtube.com) to continue the investigation.

Now in **Wireshark,** we filter the traffic and set **DNS** to see the traffic we need this time.



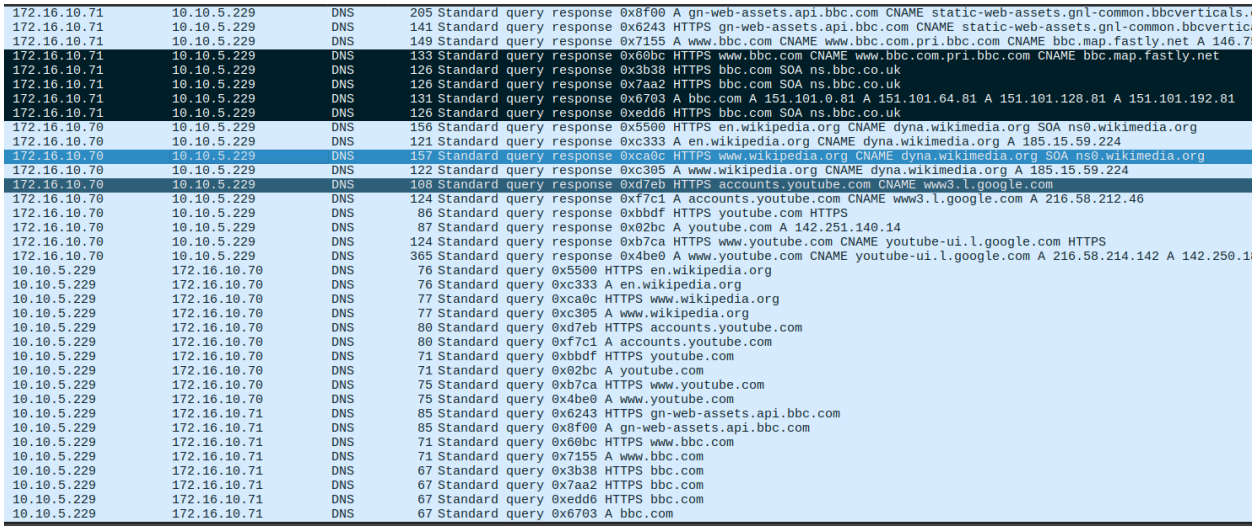
(Figure 3.1 - Wireshark filter for DNS packets**)**

In a few seconds, **Wireshark** captured 14887 packets, which are impossible to analyze manually. I used a bit more advanced filtering. I filtered only the websites I wanted to test. The filter looks like this: “dns.qry.name contains "bbc.com" or dns.qry.name contains "youtube.com" or dns.qry.name contains "wikipedia.org".



(Figure 3.2 - Filter only webpage names**)**

After filtering we got only 36 packets which is now possible to even analyze manually.



(Figure 3.3 - DNS query and response traffic**)**

The screenshot above shows **DNS query and response traffic** captured in Wireshark when accessing *bbc.com*, *wikipedia.org*, and *youtube.com*. Each line represents either a **DNS query** (a request for an IP address) or a **DNS response** (the answer from the DNS server with the corresponding IP).

- For **BBC**:
- We see queries such as **www.bbc.com** and **CNAME** records pointing to **bbc.map.fastly.net**. The response then provides multiple IP addresses (e.g., **151.101.0.81, 151.101.64.81**). This emphasizes that, even though the BBC uses a content delivery network (Fastly), DNS traffic clearly shows that the user tried to visit the BBC's website.
-
- For **Wikipedia**:

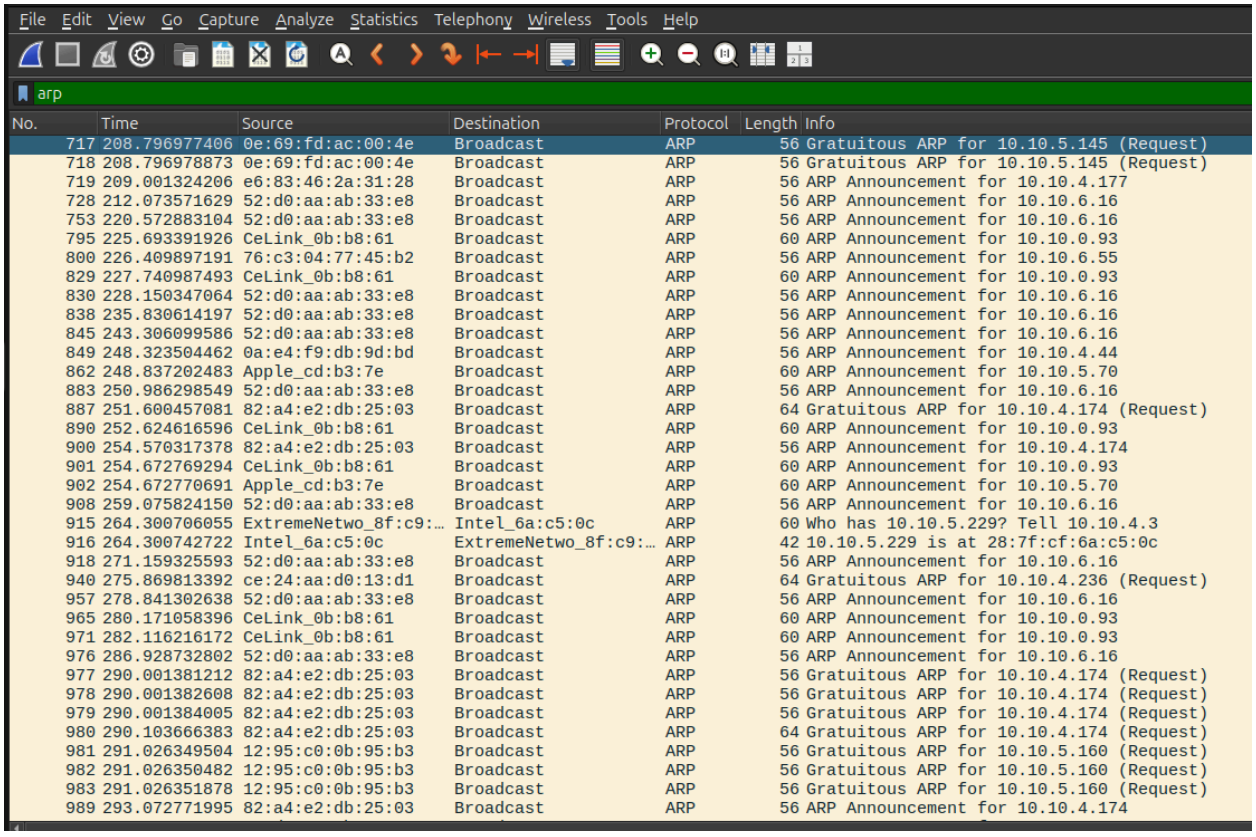
- Packets such as **A en.wikipedia.org** return IP addresses like **185.15.59.224.** We also see things on Wikimedia.org, the hosting organization for Wikipedia. Again, this shows the exact site being accessed.
-
- For **YouTube**:
- Multiple DNS queries resolve domains such as **www.youtube.com**, **accounts.youtube.com**, and **youtube-ui.l.google.com**. These point to IP addresses owned by Google (e.g.,**216.58.212.46, 142.250.140.14**). This again shows that a single visit to **YouTube** often results in multiple DNS lookups for related subdomains.

Although the actual browsing sessions are protected by **HTTPS encryption**, the DNS queries themselves are unencrypted (in this test setup)(Herzberg, 2020). This means that anyone with access to network traffic, such as an ISP, network administrator, or attacker, can still observe exactly which websites are visited.

This shows a critical vulnerability: **DNS queries leak** browsing habits, even when the content of the communication is secure. Modern mitigations, such as **DNS over HTTPS (DoH)** and **DNS over TLS (DoT)**, aim to encrypt DNS traffic(Herzberg, 2020), but in many networks, standard plaintext DNS is still used(Herzberg, 2020).

ARP Test

To continue our experiment, we again need to facilitate filtering. This time we are going to use **ARP**.

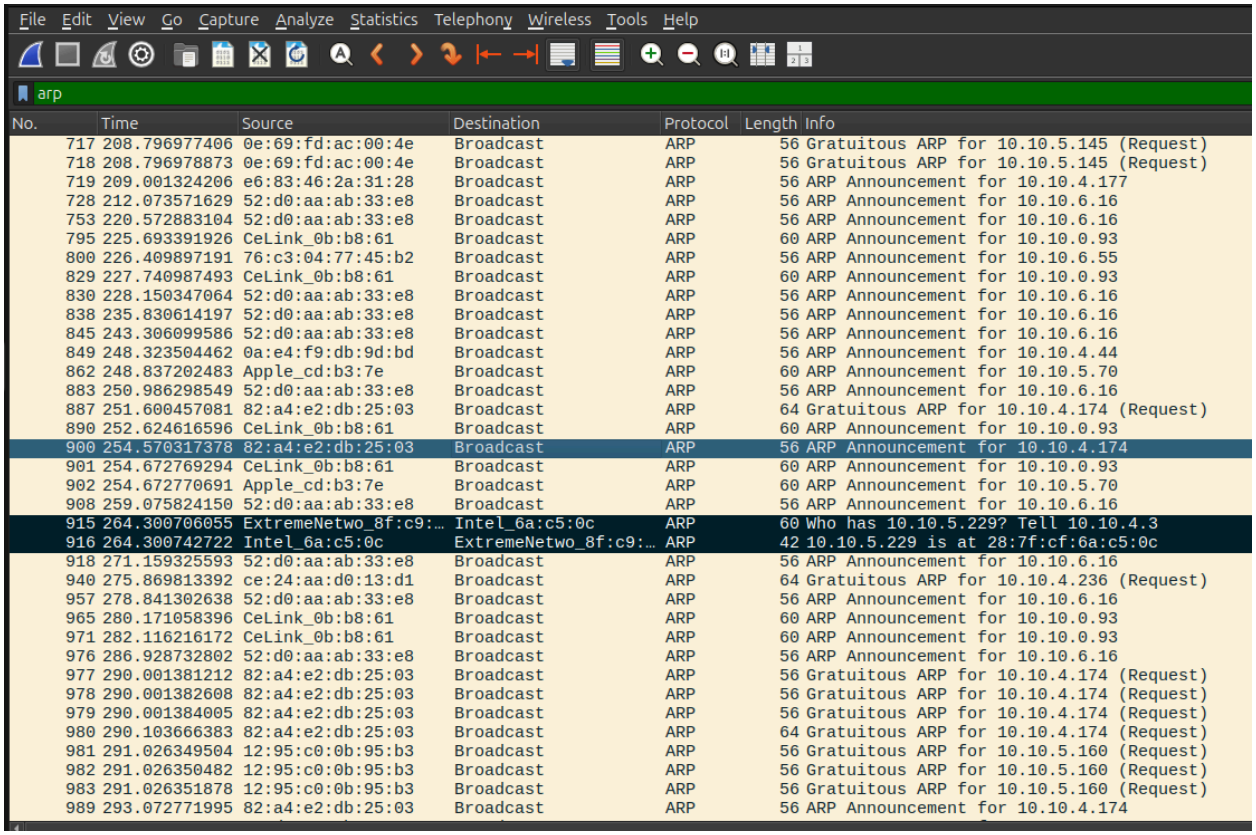


(Figure 4.1 - Wireshark with only ARP filter**)**

Figure 4.1 demonstrates the screen with only **ARP** filtered. Here we need to look for lines like:

- `Who has 192.168.1.1? Tell 192.168.1.5`

- `192.168.1.1 is at xx:xx:xx:xx:xx:xx`



(**Figure 4.2 -** Demonstrating the packets that have to be investigated)

The black-marked packets are the ones we need to continue investigating.

Packets **915** and **916** represent a fundamental exchange on any local network: the Address Resolution Protocol (**ARP**) in action(TechTarget, 2025).

Packet 915: The ARP Request

Source: ExtremeNetwo_8f:c9:f4 (IP: 10.10.4.3)

Destination: Broadcast (This is sent to every device on the local network segment)

Info: Who has 10.10.5.229? Tell 10.10.4.3

This packet is an announcement. The device at 10.10.4.3 should send data to 10.10.5.229, but it doesn't know the physical (MAC) address required to send it at the hardware level. Instead of knowing where to send the request, it screams the question to everyone on the network: "Who owns this IP address? When you find out, tell me directly at my IP address."

Packet 916: The ARP Reply

Source: Intel_6a:c5:0c (IP: 10.10.5.229)

Destination: ExtremeNetwo_8f:c9:f4 (This is sent directly to the requester)

Info: 10.10.5.229 is at 28:7f:cf:6a:c5:0c

The device with the IP address 10.10.5.229 listens for broadcasts and responds directly to the requester. It says, "I am the device you are looking for, and my MAC address is 28:7f:cf:6a:c5:0c." Device 10.10.4.3 receives this answer and adds the IP-MAC pairing to its ARP cache, ensuring it can correctly address future communication.

Why This Exchange is Necessary:

This process is the basis of local network communication. IP addresses are a logical concept for routing between networks, but real data delivery on a local Ethernet or Wi-Fi network depends on MAC addresses. ARP is the essential "translator" that bridges the logical network layer (IP) with the physical network layer (MAC) (TechTarget, 2025; Kurose & Ross, 2021)..

The Critical Vulnerability:

The big security issue with this process is that it doesn't verify that the device sending an ARP reply is actually who it claims to be(Khan, Siddiqui, & Malik, 2023). The system just trusts what it receives. This weakness allows someone malicious on the same network to launch an attack called ARP Spoofing (or ARP Poisoning) (Khan, Siddiqui, & Malik, 2023). Here's how they do it:

- The attacker listens for the ARP Request (Packet 915).
- Then, before the real device answers, the attacker sends a fake ARP Reply saying: "10.10.5.229 is at [Attacker's MAC Address]."
- The device 10.10.4.3 believes this lie and changes its ARP cache to use the attacker's address instead.
- Now, all data intended for 10.10.5.229 is sent to the attacker's machine.
- The attacker sits in the middle (Man-in-the-Middle) and can see, steal, or change all the communication between 10.10.4.3 and 10.10.5.229 without anyone noticing.
- This way, the attacker can quietly spy on or interfere with the data flowing between devices on the network.(Khan, Siddiqui, & Malik, 2023)

Analysis

Three penetration tests were made which are HTTP Login, DNS Lookup, and ARP Spoofing. Together they show a variety of vulnerabilities that typical home routers have.

The tests were done in such a way that they focus on different parts of the network, which revealed ways in which users can be exposed and how routers fail to protect them.

The HTTP Login Test exposed complete credential information in cleartext. Since HTTP traffic is transmitted unencrypted over port 80, the attacker can intercept login data using only passive listening. This shows a critical application-layer vulnerability, due to the router's inability to enforce HTTPS-only browsing or block insecure pages. This finding directly supports the research question by proving that routers in default mode are blind to encrypted traffic risks, a failure confirmed in studies by Anderson (2020) and Chai et al. (2022).

In the DNS Lookup Test, the router failed to hide or encrypt DNS queries. Although the webpage content was encrypted, the domain names visited were fully visible to passive observers. This metadata exposure allows profiling, surveillance, or phishing targeting without breaking HTTPS. Compared to HTTP sniffing, this leak is subtler but still dangerous. The absence of DNS-over-HTTPS or DNS-over-TLS features by default validates the claim that ISPs and router vendors often neglect privacy (Herzberg, 2020).

The ARP Test revealed perhaps the most dangerous vulnerability: a complete lack of verification at the data link layer. Because ARP accepts any response without validation, a malicious actor on the same network could easily perform ARP spoofing. Even though

this test did not include an active spoofing simulation, the conditions were shown to be ideal for such attacks. The router provided no detection, mitigation, or warning.

These results do not occur at the same time; they reinforce each other's impact. HTTP and DNS tests show that data can be seen, while ARP reveals how an attacker could control traffic. Together, they point to a system-wide problem, not just single issues.

Although the three tests that were done, they focused on different OSI layers, and a deeper analysis tells that all the vulnerabilities arise from the choices appearing in underlying protocols. The HTTP protocol relies on TCP and without any cryptography, which is the reason why the payload is transmitted in cleartext as it was constructed in the browser. From computer science, this shows a weakness in the protocols and not in the router. DNS uses UDP and has no authentication or guarantee that the information is confidential, which becomes an explanation why all the information is visible until DNS over HTTPS or DNS over TLS is used. ARP is even more fundamental, it accepts any reply even without authentication, so the ARP cache can be poisoned by any user. So the results reveal that the weaknesses that block router to secure users comes from not router itself, it is linked to historical designs of protocols, where were prioritized simplicity and the speed and not security.

This analysis has some limitations since it focused on just one router model and used only passive testing. Future studies could use active ARP spoofing, compare various router models with threat detection features, or test the impact of DNS encryption for clearer results.

This investigation shows that most basic home routers do not have key security features enabled by default. The essay uses different types of observations, evaluates the risks, and refers to cybersecurity research to answer the research question thoroughly.

Conclusion

This investigation showed that home routers with default settings have serious security problems. Through three tests, it became clear that private information can be captured with little effort. Passwords were sent unencrypted, visited websites were revealed, and local devices could be tricked into sending data to the wrong destination. All of this was possible using just Wireshark on a home network, without any advanced hacking tools.

These findings clearly answer the research question:Home routers are vulnerable to significant level. They do not mitigate or block common, basic threats, and many users are unaware of these risks. This emphasizes why basic network security is important, even at home. A few steps to be more secure on the internet are to turn on HTTPS-only settings, use firewall protection, and consistently update the router's firmware.

Evaluation & Limitations

This investigation showed success revealing key vulnerabilities, but at the same time it had limitations which did not allow to fully uncover the topic and examine the title. One of the main issues that occurred during the work was the restrictions set by internet providers. I tested the routers on two different Internet Provider networks, but both providers had a policy that prevented consumers from accessing admin settings. The result is that I could not enable or disable specific features such as encrypted DNS, HTTPS redirection, or ARP filtering. This has become a limit which led to my inability to compare network behavior with and without security protections, which could have strengthened the results and made them more complete.

Additionally, the tests were conducted on only one router model, which may not be fully representative of all routers, as different models have different default settings. Also, due to time and equipment constraints, the test focused solely on passive capture using Wireshark. More advanced attacks, such as spoofing simulation or real-time packet injection, were not performed, although they would reveal more severe vulnerabilities in detail, leading to a better understanding of the topic and the importance of vulnerabilities.

The decision to rely only on passive testing also influenced how I understand home network security. First I thought the HTTP login test as from my perspective it is unacceptable that it exposed full credentials in cleartext. On the other hand the ARP traffic could be intercepted which did not require any verification, this was a turning point for my focus on low level protocol designs and not only application layer configuration. Here I understood that many vulnerabilities that need to be fixed are not coming from simple user settings, because all the weaknesses and skipped security parts that were

exposed during this investigation are coming from legacy protocols such as ARP and DNS and how they were made originally. If I needed to repeat this investigation I would've combined the passive captures with active spoofing to absorb the speed of detecting or failure of detections of manipulation.

The future work must include active testing and should have ARP spoofing tests, DNS poisoning, and HTTP downgrade trials. These will let us see if the router actively detects and/or blocks the threats. This eventually would change the conclusion part. If certain attacks are mitigated, it would reveal limitations in this passive test approach and give more detailed understanding of home router security.

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